



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	2 OCTOBER 2023
GRADE	11 (2)	MARKS	40
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	40 MINUTES
CLASS			

INSTRUCTIONS:

1. This paper consists of 8 pages- including 1 page of additional writing space. Please check that your question paper is complete.
 2. Answer all questions **on the question paper**.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	10	
2	13	
3	17	
TOTAL: 40		

Question 1:

1. Select the name/ or term in Column II which best matches the description in Column I. Write the letter of the matching name/term in the appropriate space provided between **the brackets**. Each letter may only be used once. [10]

Column I

- [] Antibodies
- [] Viruses that infect bacteria
- [] Weakened immune system
- [] Algae
- [] Germs
- [] Agar
- [] Acellular
- [] Mosquito
- [] Antibiotics
- [] Thrush

Column II

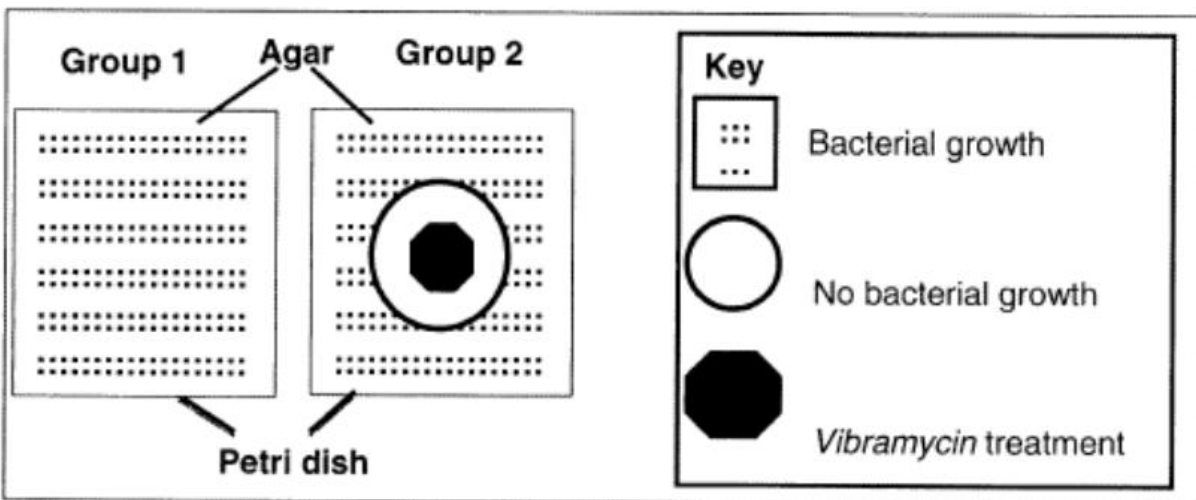
- A** Gel extract
- B** HIV
- C** Pathogens
- D** Fungus
- E** B lymphocytes
- F** Vector
- G** Protist
- H** Bacteriophage
- I** Bacteria
- J** Virus
- K** T lymphocytes
- L** decomposers

Question 2

Grade 11 learners wanted to investigate whether a new antibiotic, called Vibramycin, could restrict the production of a population of bacteria which was resistant to other antibiotics.

Their investigation was set up as follows:

- The bacteria were cultured in the same type of agar medium in 20 petri dishes of the same size and then divided into 2 groups:
 - **Group 1:** 10 petri dishes were given no treatment.
 - **Group 2:** 10 petri dishes were treated with Vibramycin of the same concentration.
- The petri dishes were then incubated under the same conditions and examined for bacterial growth.
- The diameter of the area where no bacteria grew was measured for each petri dish.



2.1. What is the dependent variable in the above investigation? (1)

2.2. Identify THREE factors that should be kept constant during the investigation, other than those mentioned above. (3)

2.3. Why is more than one petri dish used in each group? (1)

2.4. What was the aim of Group 1 in the investigation? (1)

2.5. A similar experiment was done in which the cultured bacteria were divided into 5 groups. These groups were then treated with different concentrations of Vibramycin.

a. Formulate a hypothesis for the above investigation. (2)

b. State the independent variable for this investigation. (1)

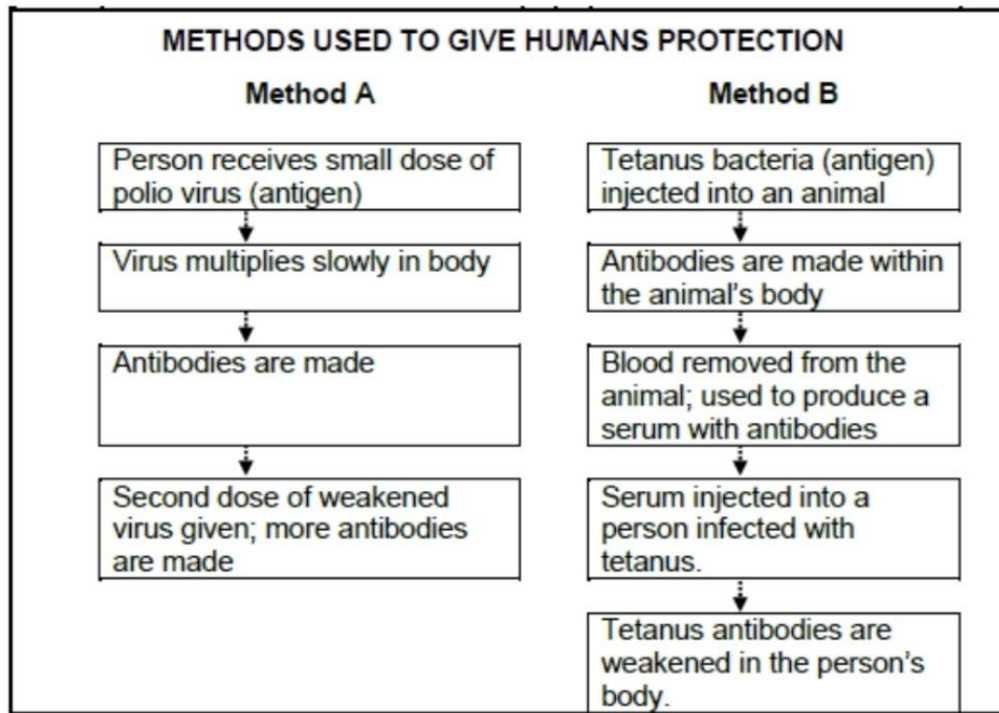
2.6. Explain how resistance to antibiotics develops in a population of bacteria. (3)

2.7. State ONE way in which the development of resistant bacteria can be minimized. (1)

[13]

Question 3

- 3.1. The diagram below shows two methods which are used to give humans protection against diseases. Method A shows active immunity and Method B shows passive immunity. Method A can be used against polio. Method B is often used against tetanus.



- 3.1.1. Name the substances produced by the body, which destroy harmful viruses and bacteria. (1)

- 3.1.2. Why does Method A give long-lasting protection against polio? (2)

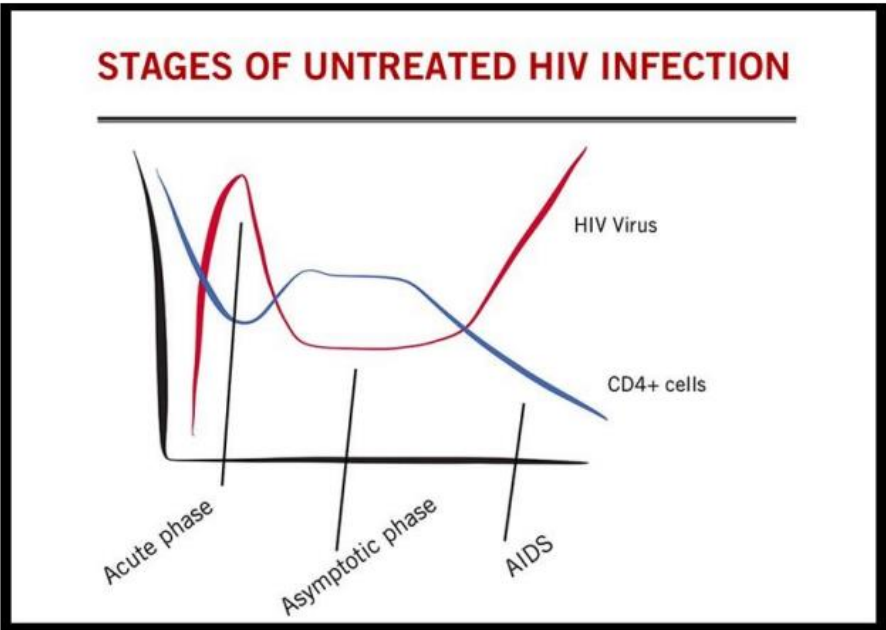
- 3.1.3. Method B does not give long-lasting protection against tetanus. Why? (2)

- 3.1.4. In immunization against polio, a second dose of the weakened virus is given (known as a booster). Suggest a reason why this booster is necessary. (2)

3.1.5. Why is Method B very good for dealing quickly with an infection like tetanus? (2)

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3.2. Acute HIV infection is the earliest stage of HIV infection, and it generally develops withing 2 to 4 weeks after infection with HIV. During this time, some people have flu-like symptoms, such as fever, headache, and rash. In the acute stage of infection, HIV multiplies rapidly and spreads throughout the body.



3.2.1. Explain the function of CD4 cells. (1)

3.2.2. When will a person be classified with AIDS? (2)

3.2.3. Explain why the HIV virus is so dangerous and causes so many deaths. (3)

3.2.4. Why is it possible to easily cultivate and grow bacteria, but it is very difficult to cultivate a virus? (2)

[8]

[17]

TOTAL: 40 MARKS

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